Zephyr Rollout Decision Analysis Timing, scope, and product coverage for this year’s firmware releases

Clinical Solution 2.0 · Limb Sensor Rev K · Aria Scratch Sensor

Decision Overview

The decision concerns the timing, scope, and product coverage of Zephyr firmware adoption during this year's firmware releases.

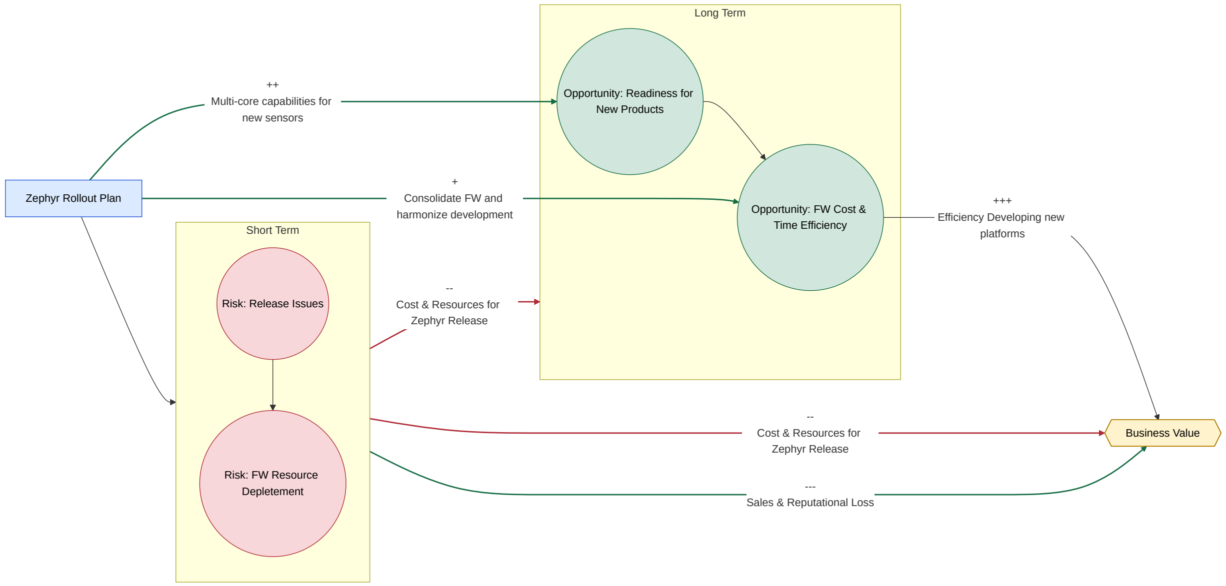
Key Releases Impacted

- **Clinical Solution 2.0** — new feature set; scope/timeline TBD (Nov release).
- **Limb Sensor Rev K (stretch goal)** — new HW revision with MFG/verification/regulatory work.
- **Aria Scratch Sensor (target)** — Q3 2026; Zephyr FW feature-complete, needs refinement.

Value & Risk Summary

- Platform reuse and tooling gains vs. regulatory/legacy constraints.
- Multi-core roadmap potential vs. added complexity and stability risk.
- Schedule, OTA/migration drag, and dual-stack maintenance risk.
- Team/new-ecosystem maturity adds execution risk.

Influence Diagram



Decision Nodes

Code	Strategy	Description
SAFE_ROLL	Conservative Rollout	No clinical Zephyr rollout this year (clinical release remains on legacy FW). Aria decision is separate.
LKWARM_ROLL	Lukewarm Rollout	Clinical Zephyr rollout on new sensor HW only (Dragonfly).
AGGR_ROLL	Aggressive Rollout	Clinical Zephyr rollout on current clinical line plus new HW (Chest RevK + Limb RevJ/K + Dragonfly as applicable).

Uncertainty Model — Short-Term Outcomes

Strategy	OK_REL	BAD_REL	Notes
SAFE_ROLL	1.0	0.0	No clinical Zephyr release so no Zephyr-driven clinical release risk in this model.
LKWARM_ROLL	0.5	0.5	Zephyr clinical release limited to Dragonfly.
AGGR_ROLL	0.3	0.7	Broadest clinical Zephyr scope (highest integration risk).

Uncertainty Model — Long-Term Readiness

Condition	READY	NOT_READY
SAFE_ROLL (no clinical Zephyr)	0.7	0.3
LKWARM_ROLL + REL_OK	0.8	0.2
LKWARM_ROLL + REL_BAD	0.2	0.8
AGGR_ROLL + REL_OK	0.8	0.2
AGGR_ROLL + REL_BAD	0.1	0.9

- SAFE_ROLL preserves roadmap focus but reduces field validation.
- REL_OK increases confidence but consumes capacity.
- REL_BAD increases drag and technical debt.

Payoff Model (Summary)

Payoff = Strategy planned cost + (release support cost if BAD_REL) + (long-term readiness value)

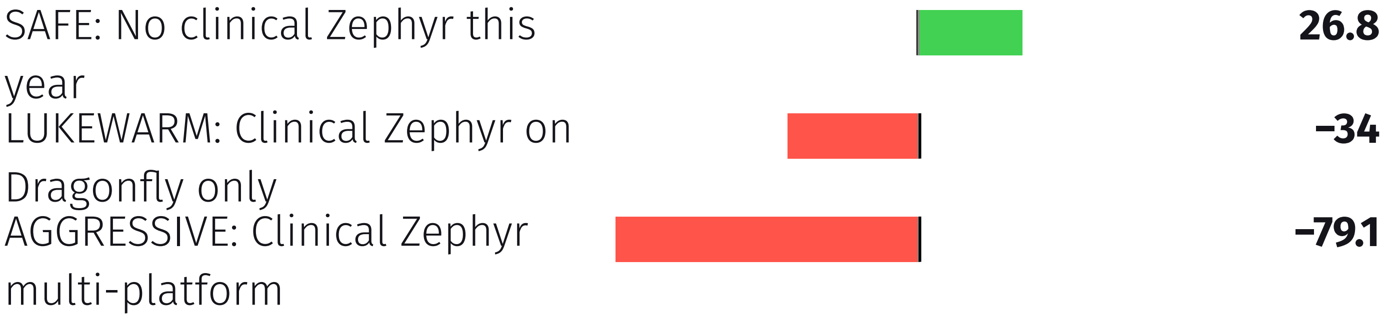
Strategy	Planned Cost	Rationale
SAFE_ROLL	−5	Minimal Zephyr clinical work; preserves capacity for long-term and maintenance.
LKWARM_ROLL	−15	Limited Zephyr clinical scope (Dragonfly only).
AGGR_ROLL	−25	Broad clinical Zephyr scope (largest verification and integration effort).

Terminal Outcome Payoffs

Strategy / Terminal Outcome	Payoff
SAFE_ROLL + READY	−5 + 0 + 50 = **45**
SAFE_ROLL + NOT_READY	−5 + 0 - 50 = **−55**
LKWARM_ROLL + OK_REL + READY	−15 + 0 + 50 = **35**
LKWARM_ROLL + OK_REL + NOT_READY	−15 + 0 - 50 = **−65**
LKWARM_ROLL + BAD_REL + READY	−15 - 40 + 50 = **−5**
LKWARM_ROLL + BAD_REL + NOT_READY	−15 - 40 - 50 = **−105**
AGGR_ROLL + OK_REL + READY	−25 + 0 + 50 = **25**
AGGR_ROLL + OK_REL + NOT_READY	−25 + 0 - 50 = **−75**
AGGR_ROLL + BAD_REL + READY	−25 - 40 + 50 = **−15**
AGGR_ROLL + BAD_REL + NOT_READY	−25 - 40 - 50 = **−115**

Expected Value Analysis

Strategy	Expected Value (pts)
AGGRESSIVE: Clinical Zephyr multi-platform	−79.1
LUKEWARM: Clinical Zephyr on Dragonfly only	−34
SAFE: No clinical Zephyr this year	26.8



Decision Tree